

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics
PRACTICAL NO - 7

Roll No.: T032	Name: KHUSHI
Class: TYIT	Batch:2
Date of Assignment: 3-10-26	Date/Time of Submission: 3-10-26

7. Sampling and Sampling Distributions:

a. Conduct a survey and collect data from a sample population, ensuring proper sampling techniques are employed.

CODE:-

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Load CSV file

```
df = pd.read_csv("Waste Management Survey Responses - Form Responses 1.csv")
```

Display first five records

```
print("First Five Records:")
print(df.head())
```

Display number of responses

```
print("\nTotal Number of Responses:", len(df))
```

Select numerical variable

```
# -----
```

#Satisfaction with waste management

```
column = "9. How satisfied are you with waste management in your area?"
satisfaction = pd.to_numeric(df[column], errors="coerce").dropna()
```

Calculate statistics

```
sample_size = len(satisfaction)
mean_value = satisfaction.mean()
median_value = satisfaction.median()
std_value = satisfaction.std()
print("\nStatistics:")
print("Sample Size:", sample_size)
print("Mean:", round(mean_value, 2))
print("Median:", round(median_value, 2))
print("Standard Deviation:", round(std_value, 2))
```

Display frequency of satisfaction ratings

```
print("\nSatisfaction Frequency:")
print(satisfaction.value_counts().sort_index())
```

Create suitable chart

```
plt.figure(figsize=(7,5))
satisfaction.value_counts().sort_index().plot(
    kind="bar"
)
plt.title("Waste Management Satisfaction Survey")
plt.xlabel("Satisfaction Rating")
plt.ylabel("Number of Respondents")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

```
print("Khushi T032")
```

OUTPUT:-

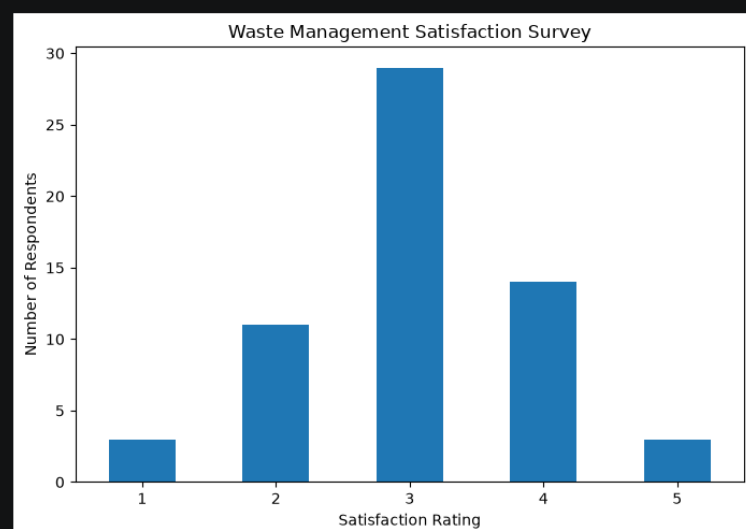
```
First Five Records:
      Timestamp      1. Name \
0  9/19/2026 22:30:54      Azma
1  9/19/2026 22:45:29  Leena Mondal
2  9/19/2026 22:55:44 Santosh prajapati
3  9/19/2026 23:08:32      Radhika
4  9/19/2026 23:14:59      Sid

2. Which area/locality do you live in? \
0      Santacruz (east)
1      Bhayandar
2      Naigaon east
3      Andheri East
4  Santacruz/ Andheri

3. How many people are there in your household? \
0      52
1      4
2      180
3      4
4      3
```

```
4. Approximately how much waste does your household generate per day? (in kg) \
0      3/4 kg
1      1kg
...
3      29
4      14
5      3
Name: count, dtype: int64
```

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b. Use the Central Limit Theorem to analyze the sampling distribution of a sample mean and estimate population parameters.

CODE:-

Select random sample of 30 observations

```
np.random.seed(42)
sample_30 = satisfaction.sample(
    n=30,
    random_state=42 )
print("\nRandom Sample of 30:")
print(sample_30)
```

Calculate mean of selected sample

```
sample_mean = sample_30.mean()
print("\nMean of Selected Sample:", round(sample_mean, 2))
```

Repeat sampling 100 times

```
sample_means = []
for i in range(100):
    random_sample = satisfaction.sample(
        n=30,
        replace=False
    )
    mean = random_sample.mean()
    sample_means.append(mean)
```

Store sample means

```
sample_means_df = pd.DataFrame({
    "Sample Number": range(1, 101),
    "Sample Mean": sample_means
})
```

Display first 10 sample means

```
print("\nFirst 10 Sample Means:")
print(sample_means_df.head(10))
```

Mean of all sample means

```
mean_of_sample_means = np.mean(sample_means)
print(
    "\nMean of All Sample Means:",
    round(mean_of_sample_means, 3)
)
```

Standard deviation of sample means

```
sd_sample_means = np.std(
    sample_means,
    ddof=1
)
print(
    "Standard Deviation of Sample Means:",
    round(sd_sample_means, 3)
)
```

Histogram of sampling distribution

```
plt.figure(figsize=(7,5))
```

```
plt.hist(
    sample_means,
    bins=10,
    edgecolor="black"
)
plt.title(
    "Sampling Distribution of Sample Means (n = 30)"
)
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```

Standard Error

```
standard_error = std_value / np.sqrt(30)
print(
    "\nStandard Error:",
    round(standard_error, 3)
)
```

Compare original mean and sampling mean

```
print("\nComparison:")
print(
    "Original Sample Mean:",
    round(mean_value, 3)
)
print(
    "Mean of Sampling Distribution:",
    round(mean_of_sample_means, 3)
)
```

Sampling distributions for different sizes

```
sample_sizes = [10, 20, 30, 40]
results = []
for n in sample_sizes:
    means = []
    for i in range(100):
        random_sample = satisfaction.sample(
            n=n,
            replace=False
        )
        means.append(random_sample.mean())
    results.append({
        "Sample Size": n,
        "Mean of Sample Means": np.mean(means),
        "SD of Sample Means": np.std(
            means,
            ddof=1
        )
    })
comparison = pd.DataFrame(results)
print("\nComparison of Different Sample Sizes:")
print(comparison.round(3))
```

Estimate population mean

```
population_mean_estimate = mean_value
print(
    "\nEstimated Population Mean:",
    round(population_mean_estimate, 2)
)
```

Save sample means to CSV

```
sample_means_df.to_csv(
    "Sampling_Distribution_Sample_Means.csv",
    index=False
)
print(
    "\nSample means saved successfully!"
)
```

Conclusion

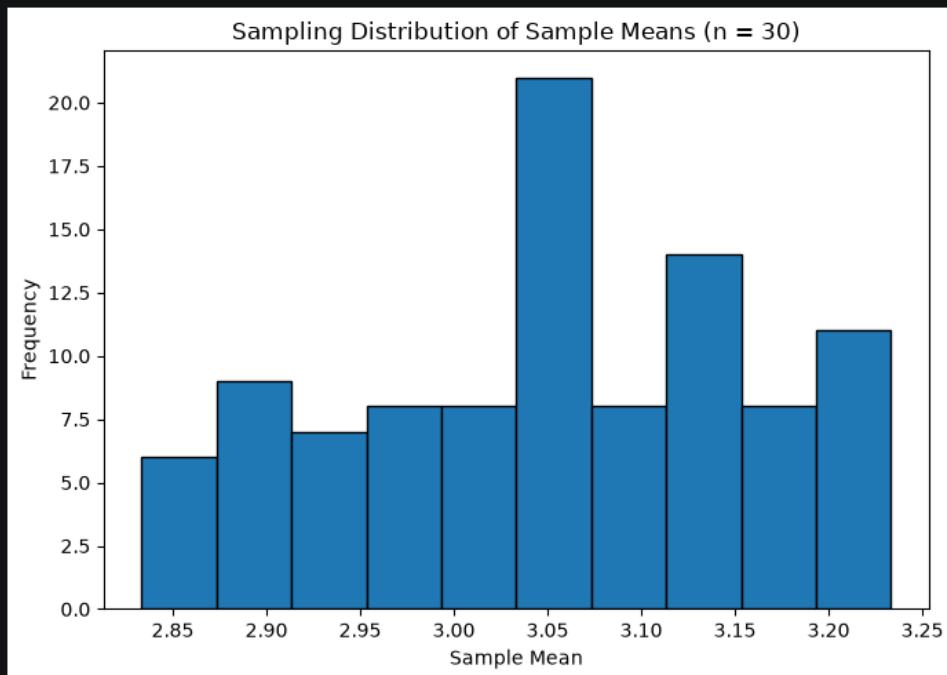
```
print("\nConclusion:")
print(
    "The sampling distribution of the sample means is centered "
    "close to the original sample mean. As the sample size "
    "increases, the variation among sample means decreases. "
    "This demonstrates the Central Limit Theorem and shows "
    "that the sample mean can be used to estimate the "
    "population mean."
)
print("Khushi T032")
```

OUTPUT:-

```
Random Sample of 30:
0      2
5      4
36     5
45     3
13     2
54     1
33     3
48     4
12     3
57     2
46     3
50     3
31     3
3      3
52     3
17     2
8      4
6      3
40     4
4      3
43     1
19     2
34     4
...
9      10      2.900000

Mean of All Sample Means: 3.049
Standard Deviation of Sample Means: 0.105
```

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Standard Error: 0.166

Comparison:

Original Sample Mean: 3.05

Mean of Sampling Distribution: 3.049

Comparison of Different Sample Sizes:

	Sample Size	Mean of Sample Means	SD of Sample Means
0	10	3.052	0.255
1	20	3.046	0.160
2	30	3.049	0.130
3	40	3.042	0.080

Estimated Population Mean: 3.05

Sample means saved successfully!

Conclusion:

The sampling distribution of the sample means is centered close to the original sample mean. As the sample size in
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